Visual Analytics
Fall 2025 updated on 16.12.2025

Exam structure and rules

(read these slides carefully)

Exam

Exam encompasses 3 main parts

- **Project (mandatory)** **up to25 points**
  - **a full working visual analytics solution**
- **Report (mandatory)** **up to 5 points**
  - **a up to 5 pages report structured as a scientific paper describing your project and <u>showing proposals close to your solution</u>**
- **October home_work (optional)** **up to3 points**
  - **It will be evaluated at exam time**

Rules

The exam can be completed **individually** or in **group** of MAX 3 persons.

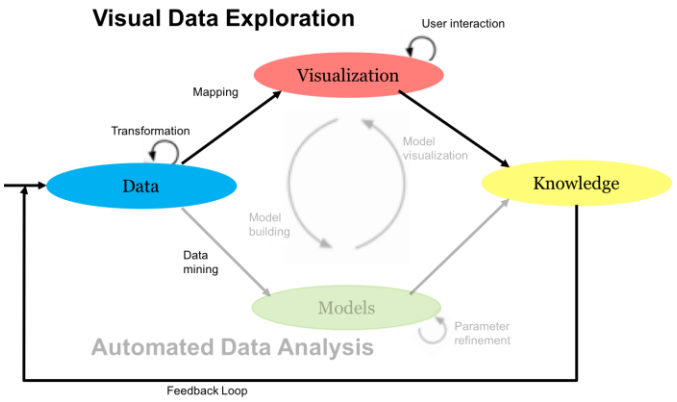
2 tiers of projects:

- **Free proposal**
  - Project proposal comes from the student/group (select data that you like!!)

- **Project proposed by me (more challenging projects, group of 2/3 people):**
  - We will provide a specification for the project (dataset, goals, eventual additional requirements) on demand
  - You need to write me an e-mail explicitly requiring to conduct this tier
  - The proposal will be positioned on themes of my research group
  - If the work is of good quality, it can be continued/extended along with the Master Thesis

Rules

- For both tiers**, a preliminary approval is needed**: before starting working on the project you **MUST** send me an e-mail with a **1 page draft** of the idea that:
  - Specify the dataset (characteristics and context)
  - Specify the general idea (Analytics part, Visual Part)
    1. Data
    2. Used visualizations and dimensionality reduction
    3. Used analytics
    4. Coordinated views (a least 2)
    5. How do you trigger **visually** the analytics
       - Not using a menu, not selecting a radio button
    6. How the analytics result is presented to the user
  - Specify the intended user
  - A mockup of the user interface (draft)



Rules

- **Mandatory**: the project MUST use a dimensionality reduction technique (e.g.,PCA, MDS, or a t-SNE) and integrate it in the analysis flow

- **Mandatory**: **VISUALIZATION:** every assignment **MUST** have a visual part constituted by at least 2 visualizations coordinated in both ways and interactive

- **Mandatory** **ANALYTICS**: every assignment MUST contain at least 1 computation **that is triggered by user visual interactions** (no change in the dataset, no simple filter, no simple selection from a menu)

Project positioning

- Analysis of scientific papers related to the project
  - **similar** technical solution
  - **similar** objectives
  - **similar** dataset

- Discussion of:
  - related proposals
  - related techniques
  - **differences** with respect to the assignment

Project positioning

Where to look for related proposals:

Text search

- Google Scholar: https://scholar.google.it/schhp?hl=it
- IEEE Explore: https://ieeexplore.ieee.org/Xplore/home.jsp
- ACM Digital Library: https://dl.acm.org/

Search by topics (e.g. , visualization for network medicine) and browse the results

Scientific related venues:

- IEEE VIS
- EuroVIS

Browse the proceedings (look at the papers of each edition and select the most related ones to your exam)

Dataset

You are all encouraged to **propose** a context and a datasets of your choice on topics that you find interesting to analyze, e.g.:

1. Sport analytics (football, tennis, sailing, etc.)
2. Bitcoin, block chain
3. Network traffic data
4. Social networks analysis
5. Vast 202x challenge
6. Data coming from the datasets described in the following slides

- The Dataset **MUST** respect the rule that the index AS (AngeliniSantucci) defined as:

AS= #tuples * #dimensions

is contained in the range 10,000 – 50,000 (and more for the braves…..)

Datasets

DataSets:

1. **Data from ISTAT:**
   - Many different datasets are available from a main data-warehouse

http://dati-censimentoindustriaeservizi.istat.it/

2. **Italian Government Open Data**

https://www.dati.gov.it/view-dataset

3. **Top World Universities Rankings**:

http://www.topuniversities.com/university-rankings

Datasets

DataSets:

4. **IEEE VAST-Challenge in the last years** (Conference on Visual Analytics Science and Technology)
   - 3 different mini-challenge, application for 1 of them
   - An e-mail address is required for downloading the dataset
   - Check for last edition when it get published (still not, usually January)
   - Possibility to submit a work on the challenge if good (coordination needed with us, write an e-mail for that)

https://vast-challenge.github.io/2025/

https://vast-challenge.github.io/2024/ ...

5. **Users Ratings on Movies**:
   - Also IMDB can be a good resources for well-known ones.

http://grouplens.org/datasets/movielens/

6. **UCI datasets**:

https://archive.ics.uci.edu/ml/index.php

Datasets

DataSets:

7. **Italian Top Scientists**:

http://www.topitalianscientists.org/

8. **Kaggle datasets**:

https://www.kaggle.com/datasets

9. **CyberSecurity datasets**:

http://vizsec.org/data/

10. **Network analysis (generic networks) data**:

http://snap.stanford.edu/

11 **COVID-19 data**:

https://github.com/pcm-dpc/COVID-19

Project structure

- The assignment **MUST** contain (you have to provide a github link):
    - The running **software**
    - A relation (5-6 pages) **similar to a scientific paper** that describes the whole design process, rationale and prototype, comprehensive of the **related work** section that describes **the related papers**, and **the discovered insights** section.
    - A PowerPoint **presentation** describing the goal of the project, the data structure, and the chosen visualizations (**not the user interaction**, that must be presented during the demo of the system)
- The exam will consist in
    - a) presenting the PowerPoint material (each student MUST present a part of the content) max 20'
    - b) giving a live demo of the system
    - c) answering questions on both the presentation and the demo

Steps and dates

1. Create a **group**
2. Select a **dataset**, the domain of the VA application, and the intended user(s)
3. Send me **1 page draft for the approval** request (when you have it ready)
   - In the email subject write **VA 2025project approval request**
4. Wait for the approval and the group number 25_x (use it in any following mail messages)
5. **Exam booking**: (exams will be in presence and remotely)
   - **January and February 2026**: when you are ready send an email to me by 10 January or by 10 February with the github link
     - In the email subject write **group 25_x exam booking**
     - The exam will take place in the next 10 / 15 days
   - Other sessions: send me an email by the first date of the session with the github link
     - In the email subject write **group 25_x exam booking**
     - The exam will take place in the next 10 / 15 days
   - Contact me for emergencies (Erasmus, Master degree exam approaching, etc.)
     - In the email subject write **group 25_x** <request>

Evaluation criteria

- **Create projects that are useful:**
  - Target one or more potential users of your system
  - Use your system to find insights, not just for interacting with it
  - Refine your project to reach the desired insights

- **Prioritize quality over quantity:**
  - A well-crafted system composed of 3/4 refined visualizations, highly coordinated and interactive, and with good analytical functions is BETTER than a 10 visualizations dashboard with limited interaction and no implemented analytics

- **Avoid Pitfalls**:
  - see next slide

Exclusion criteria and penalty list

- Exclusion criteria (the project will not be evaluated)
  - Missing or wrong 1 page draft of the idea
  - Missing dimensionality reduction
  - Missing GitHub site

- Penalty list
  - Missing two coordinated visualizations (5 points penalty)
  - Missing related work (5 points penalty)
  - Missing analytics triggered by **visual** interaction (5 points penalty)
  - Dimensionality reduction NOT integrated in the analysis flow (2 points penalty)
  - Not standard color encodings (2 points penalty)
  - Missing legends (2 points penalty)
  - Scrollable views not justified by design constraints (2 points penalty)
  - Too strong reuse of existing solutions (2 points penalty)
  - Missing insights (2 points penalty)
  - Missing or not convincing intended system user (2 points penalty)